



AI-BASED SMART GREENHOUSE AUTOMATION PLATFORM



A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

The AI-Based Smart Greenhouse Automation Platform is a smart greenhouse monitoring and automation system developed using ESP32, environmental sensors, IoT, and AI-based recommendations. The system monitors important greenhouse parameters such as temperature, humidity, rain conditions, light intensity, and air quality using DHT22, rain sensor, LDR, and MQ-135 sensors. The ESP32 collects sensor readings and sends the real-time values to Blynk IoT platform. The Blynk dashboard provides the user with the current greenhouse conditions and the operating status of connected devices. Based on monitored conditions, the system provides automatic control the water pump and 12 V DC fan through a relay module. The system also provides a manual mode, allowing the user to operate the devices manually when required. The AI component of the proposed platform uses the sensor output values available through the Blynk dashboard to provide meaningful suggestions to user. For ex., based on high temperature, humidity, rain, light, or air-quality readings, AI can suggest actions such as increasing ventilation, avoiding unnecessary irrigation, or monitoring the greenhouse condition. Thus, AI acts as a decision-support and recommendation system, without using a separate machine-learning model. The proposed platform provides real-time monitoring, automatic and manual device control, The system offers a low-cost, practical, and expandable solution for smart greenhouse management, helping reduce manual intervention and supporting better environmental management for plant growth.

Keywords: Smart Greenhouse, IoT, ESP32, Blynk, Greenhouse Automation, Environmental Monitoring, DHT22, LDR, Rain Sensor, MQ-135, Automatic Irrigation, Fan Control, AI-Based Recommendations.

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LIST OF ABBREVIATIONS

ESP32	Espressif 32-bit Microcontroller
IoT	Internet of Things
AI	Artificial Intelligence
DHT22	Digital Temperature and Humidity Sensor
LDR	Light Dependent Resistor
MQ-135	Metal Oxide Semiconductor Gas Sensor
DC	Direct Current
Wi-Fi	Wireless Fidelity
RH	Relative Humidity
GND	Ground

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF SMART AGRICULTURE

Agriculture plays an important role in food production and economic development. Conventional agricultural practices depend largely on manual monitoring and control of environmental conditions. Factors such as temperature, humidity, rainfall, light intensity, and air quality have a significant influence on plant growth and crop productivity. Continuous manual monitoring of these parameters can be difficult, especially when environmental conditions change rapidly.

Smart agriculture uses modern technologies such as sensors, microcontrollers, Internet of Things (IoT), automation, and Artificial Intelligence (AI) to improve agricultural management. Sensors can collect environmental information in real time, while IoT technology allows the collected data to be monitored remotely. Automation can reduce manual intervention by controlling agricultural equipment according to predefined conditions.

The proposed project applies these technologies to greenhouse management. An ESP32 microcontroller is used to collect sensor data and communicate with the Blynk IoT platform through Wi-Fi. The system monitors important environmental parameters and provides automatic control of selected greenhouse devices.

1.2 SMART GREENHOUSE SYSTEM

A greenhouse provides a controlled environment for growing plants by regulating important environmental conditions. However, maintaining suitable conditions manually requires continuous observation and operation of different devices. A smart greenhouse system uses sensors and automation technology to continuously monitor environmental parameters. In the proposed system, a DHT22 sensor is used to measure

temperature and humidity, a rain sensor detects rain conditions, an LDR module monitors light intensity, and an MQ-135 sensor monitors air-quality-related conditions.

The ESP32 acts as the main controller and processes the sensor readings. The readings are transmitted to the Blynk IoT platform through Wi-Fi, where the user can monitor the greenhouse conditions in real time. A relay module is used to control the 12 V DC fan and water pump. The system supports both automatic and manual operation.

1.3 CHALLENGES IN EXISTING GREENHOUSE MANAGEMENT

Traditional greenhouse management involves several challenges. The major challenge is the requirement for continuous human monitoring of environmental conditions. Changes in temperature and humidity may not always be identified immediately, which can affect plant growth.

Manual operation of water pumps can also result in unnecessary water consumption when irrigation is performed without considering rain or other environmental conditions. Similarly, controlling ventilation manually may not provide a quick response to high-temperature conditions.

Monitoring light intensity and air quality manually is also difficult because these parameters can change throughout the day. In addition, conventional systems may provide sensor readings without giving meaningful recommendations to the user. These limitations create a need for an integrated monitoring and automation system.

1.4 NEED FOR INTELLIGENT GREENHOUSE MONITORING

Intelligent greenhouse monitoring is required to maintain suitable environmental conditions while reducing manual intervention. Continuous monitoring allows users to identify changes in greenhouse conditions at an early stage.

An IoT-based system provides the advantage of remote monitoring, allowing

users to view sensor values and device status through a digital dashboard. Automatic control can operate devices such as fans and water pumps based on predefined conditions.

The addition of AI-based recommendations can further assist the user by interpreting available sensor values and suggesting suitable actions. For example, high temperature may result in a recommendation to improve ventilation, while rain conditions may indicate that unnecessary irrigation should be avoided.

1.5 PROPOSED SYSTEM OVERVIEW

The proposed AI-Based Smart Greenhouse Automation Platform is designed to monitor and control important greenhouse environmental conditions using ESP32, sensors, IoT, and AI-based recommendations.

The DHT22 sensor measures temperature and humidity, while the rain sensor detects rain conditions. The LDR module monitors light intensity and the MQ-135 sensor monitors air-quality-related conditions. The ESP32 receives these sensor values and sends the data to the Blynk IoT platform through Wi-Fi.

Based on predefined conditions, the ESP32 controls the water pump and 12 V DC fan through a relay module. The system also provides a manual mode through which the user can control the connected devices when required.

The AI component uses the sensor values available through the Blynk dashboard to provide recommendations to the user. It acts as a decision-support system rather than using a separate machine-learning model. Thus, the proposed system combines environmental monitoring, automatic control, manual control, remote IoT monitoring, and AI-assisted recommendations.

1.6 SCOPE OF THE PROJECT

The scope of the project is to develop a low-cost and practical smart greenhouse monitoring and automation platform. The system focuses on monitoring temperature, humidity, rain conditions, light intensity, and air-quality-related parameters.

The ESP32 is used as the main controller, while the Blynk platform provides real-time remote monitoring. The water pump and 12 V DC fan are controlled automatically using a relay module according to predefined environmental conditions.

The project also includes a manual operating mode and AI-based recommendations using the available sensor values. The system can be further expanded by adding additional sensors, actuators, cloud data storage, advanced prediction methods, and more intelligent control algorithms.

1.7 OBJECTIVES OF THE PROJECT

The main objectives of the proposed project are:

- To monitor greenhouse temperature and humidity using the DHT22 sensor.
- To detect rain conditions using a rain sensor.
- To monitor light intensity using an LDR module.
- To monitor air-quality-related conditions using the MQ-135 sensor.
- To automatically control the water pump according to predefined conditions.
- To automatically control the 12 V DC fan for greenhouse ventilation.
- To provide manual control of connected devices when required.
- To provide real-time greenhouse monitoring through the Blynk IoT platform.
- To provide AI-based suggestions using the available sensor values.
- To reduce manual intervention in greenhouse management.

1.8 NEED FOR THE PROJECT

The project is needed to overcome the limitations of conventional greenhouse monitoring and manual device operation. Continuous manual observation requires time and effort and may not provide immediate responses to changing environmental conditions.

An automated system can continuously collect environmental data and operate connected devices according to predefined conditions. This can help maintain a more suitable greenhouse environment and avoid unnecessary operation of the water pump and fan. The IoT-based monitoring facility allows users to observe greenhouse conditions remotely. AI-based recommendations provide additional decision support by converting sensor values into useful suggestions. Therefore, the project provides an economical and expandable approach to smart greenhouse management.

CHAPTER 2

LITERATURE REVIEW

Various studies have been carried out to improve agricultural productivity and greenhouse management using technologies such as IoT, environmental sensors, embedded systems, automation, and Artificial Intelligence. Existing research focuses on environmental monitoring, automatic irrigation, temperature and humidity control, and remote monitoring of greenhouse conditions. However, many systems focus on individual parameters or provide automation without intelligent recommendations. Therefore, an integrated system that combines multi-parameter monitoring, automatic device control, IoT-based remote monitoring, manual operation, and AI-based suggestions is required. The proposed system utilizes ESP32, environmental sensors, relay-controlled devices, Blynk IoT, and AI-based recommendations to provide an efficient smart greenhouse management solution.

IoT-Based Smart Greenhouse Monitoring System

Paper Title: IoT-Based Smart Greenhouse Monitoring and Control System

Authors: A. Kumar, S. Raj

This research presents an IoT-based greenhouse monitoring system for measuring environmental parameters such as temperature, humidity, and soil conditions. Sensors are connected to a microcontroller, and the collected data is transmitted to an online platform for remote monitoring. The system reduces the need for continuous manual observation. However, the system mainly focuses on monitoring and does not provide extensive AI-based recommendations or flexible manual and automatic device control.

Automated Greenhouse Temperature and Humidity Control

Paper Title: Automated Greenhouse Environment Control Using Sensors and Microcontroller

Authors: R. Sharma, P. Singh

The authors proposed an automated greenhouse system that monitors temperature and humidity and controls ventilation equipment according to predefined threshold values. The system helps maintain suitable environmental conditions for plant growth. However, the system concentrates mainly on temperature and humidity and does not consider other parameters such as rain, light intensity, and air quality.

IoT-Based Automatic Irrigation System

Paper Title: IoT-Based Smart Irrigation and Greenhouse Automation System

Authors: K. Ramesh, M. Anitha

This study presents an IoT-based irrigation system that automatically operates a water pump according to environmental conditions. Remote monitoring is provided through an IoT platform, reducing unnecessary manual intervention and improving water management. However, the system mainly concentrates on irrigation and does not provide integrated monitoring of light intensity and air quality or AI-based recommendations.

Smart Greenhouse Monitoring Using ESP32

Paper Title: Smart Greenhouse Monitoring and Automation Using ESP32

Authors: P. Kumar, V. Raj

This research uses an ESP32 microcontroller with environmental sensors to monitor greenhouse conditions. The ESP32 provides wireless connectivity and allows sensor information to be transmitted to an IoT platform. The system offers a low-cost solution for greenhouse monitoring and automation. However, the

system may be limited to predefined automation rules and does not provide intelligent suggestions based on multiple sensor values.

AI-Based Smart Agriculture System

Paper Title: Artificial Intelligence-Based Decision Support System for Smart Agriculture

Authors: S. Prakash, R. Mohan

The authors proposed an AI-based agricultural decision-support system that uses collected environmental data to provide recommendations for agricultural activities. AI can help users interpret sensor information and make suitable decisions. However, such systems may require large amounts of data and advanced machine-learning models, increasing system complexity. The proposed project uses AI as a recommendation and decision-support component based on available Blynk sensor values, without requiring a separate machine-learning model.

Research Gap and Motivation

From the literature review, it is observed that existing greenhouse systems commonly focus on environmental monitoring, irrigation automation, temperature control, or IoT connectivity individually. Some systems provide automatic control, while others focus mainly on displaying sensor readings. There is a need for a simple and integrated system that combines temperature and humidity monitoring, rain detection, light intensity monitoring, air-quality monitoring, automatic fan and water pump control, manual operation, real-time Blynk monitoring, and AI-based recommendations.

The proposed AI-Based Smart Greenhouse Automation Platform addresses this gap by integrating multiple environmental sensors with an ESP32 microcontroller and Blynk IoT platform. The system provides both automatic and manual control of the fan and water pump, while AI-based recommendations

assist the user in understanding the monitored conditions and taking appropriate actions. This integrated approach provides a low-cost, practical, and expandable solution for smart greenhouse management.

CHAPTER 3

PROBLEM IDENTIFICATION

3.1 PROBLEM IDENTIFIED

Greenhouse cultivation requires proper monitoring and control of environmental conditions for healthy plant growth. In conventional greenhouse systems, environmental parameters are often monitored manually, and devices such as fans and water pumps are operated based on human observation. This can lead to delayed responses, unnecessary resource consumption, and difficulty in maintaining suitable conditions. The major problems identified in the existing greenhouse management system are described below.

3.1.1 Continuous Manual Monitoring of Greenhouse Conditions

Continuous manual monitoring of greenhouse conditions requires regular observation of temperature, humidity, rain, light intensity, and air quality. This increases the workload of the user and may result in delayed action when environmental conditions change suddenly.

3.1.2 Uncontrolled Temperature and Humidity

Temperature and humidity directly influence plant growth. Without continuous monitoring and timely control, excessive temperature or unsuitable humidity conditions may affect the greenhouse environment and plant development.

3.1.3 Unnecessary Water Pump Operation

Manual operation of the water pump may result in unnecessary irrigation. For example, operating the pump during or after rainfall can lead to excessive water usage. An automated system is therefore required to control the pump according to environmental conditions.

3.1.4 Absence of Real-Time Greenhouse Monitoring

Traditional greenhouse management may not provide real-time information about environmental conditions. Users may have to be physically present near the greenhouse to observe sensor readings and device status.

3.1.5 Difficulty in Monitoring Light and Air Quality

Light intensity and air quality can change according to environmental conditions and time of day. Continuous manual monitoring of these parameters is difficult and may prevent the user from responding quickly to undesirable conditions.

3.1.6 Lack of Intelligent Suggestions Based on Sensor Values

Basic monitoring and automation systems generally display sensor readings or operate devices using predefined conditions. They may not provide meaningful suggestions to help users understand the collected data and decide suitable actions for greenhouse management.

3.2 PROPOSED SOLUTION

The proposed AI-Based Smart Greenhouse Automation Platform provides an integrated solution for monitoring and controlling greenhouse environmental conditions. The system uses an ESP32 microcontroller as the main controller and connects multiple sensors to monitor temperature, humidity, rain, light intensity, and air-quality-related conditions.

The DHT22 sensor is used to measure temperature and humidity, while the rain sensor detects rain conditions. The LDR module monitors light intensity, and the MQ-135 sensor monitors air-quality-related conditions. The ESP32 collects the sensor readings and sends the data to the Blynk IoT platform through Wi-Fi.

Based on predefined conditions, the ESP32 controls the 12 V DC fan and water pump through a 4-channel relay module. The automatic mode enables the system to operate the connected devices according to the monitored conditions. A manual mode is also provided so that the user can control the devices when required.

The AI component uses the sensor values available through the Blynk dashboard to provide useful recommendations. For example, high temperature can result in a suggestion to improve ventilation, while detected rain can result in a recommendation to avoid unnecessary irrigation. The AI therefore acts as a decision-support and recommendation system rather than a separate machine-learning model.

The proposed system provides real-time monitoring, automatic control, manual control, and AI-assisted recommendations in a single platform. This helps reduce manual intervention and provides a practical and expandable approach to greenhouse management.

3.3 OBJECTIVES

3.3.1 Temperature and Humidity Monitoring

To continuously monitor greenhouse temperature and humidity using the DHT22 sensor and provide the measured values through the IoT dashboard.

3.3.2 Rain Condition Detection

To detect the presence of rain using a rain sensor and use the detected condition for suitable greenhouse management and water pump control.

3.3.3 Light Intensity Monitoring

To monitor the light intensity inside the greenhouse using an LDR module and provide the corresponding sensor values to the user.

3.3.4 Air Quality Monitoring

To monitor air-quality-related conditions using the MQ-135 sensor and provide the readings for greenhouse environmental assessment.

3.3.5 Automatic Water Pump Control

To automatically control the water pump through a relay according to predefined environmental conditions and reduce unnecessary irrigation.

3.3.6 Automatic Fan Control

To automatically control the 12 V DC fan through a relay based on temperature and other predefined conditions to support proper greenhouse ventilation.

3.3.7 Manual Mode for Device Control

To provide a manual operating mode that allows the user to control the water pump and fan whenever manual operation is required.

3.3.8 Real-Time Monitoring

To provide real-time monitoring of greenhouse sensor values and device status through the Blynk IoT platform using Wi-Fi connectivity.

3.3.9 AI-Based Suggestions Using Blynk Values

To provide useful AI-based recommendations by interpreting the sensor values available through the Blynk dashboard and assisting the user in making suitable decisions.

3.3.10 Reduction of Manual Intervention in Greenhouse Management

To reduce the need for continuous manual monitoring and device operation by integrating sensing, IoT monitoring, automatic control, and AI-assisted recommendations into a single system.

CHAPTER 4

METHODOLOGY

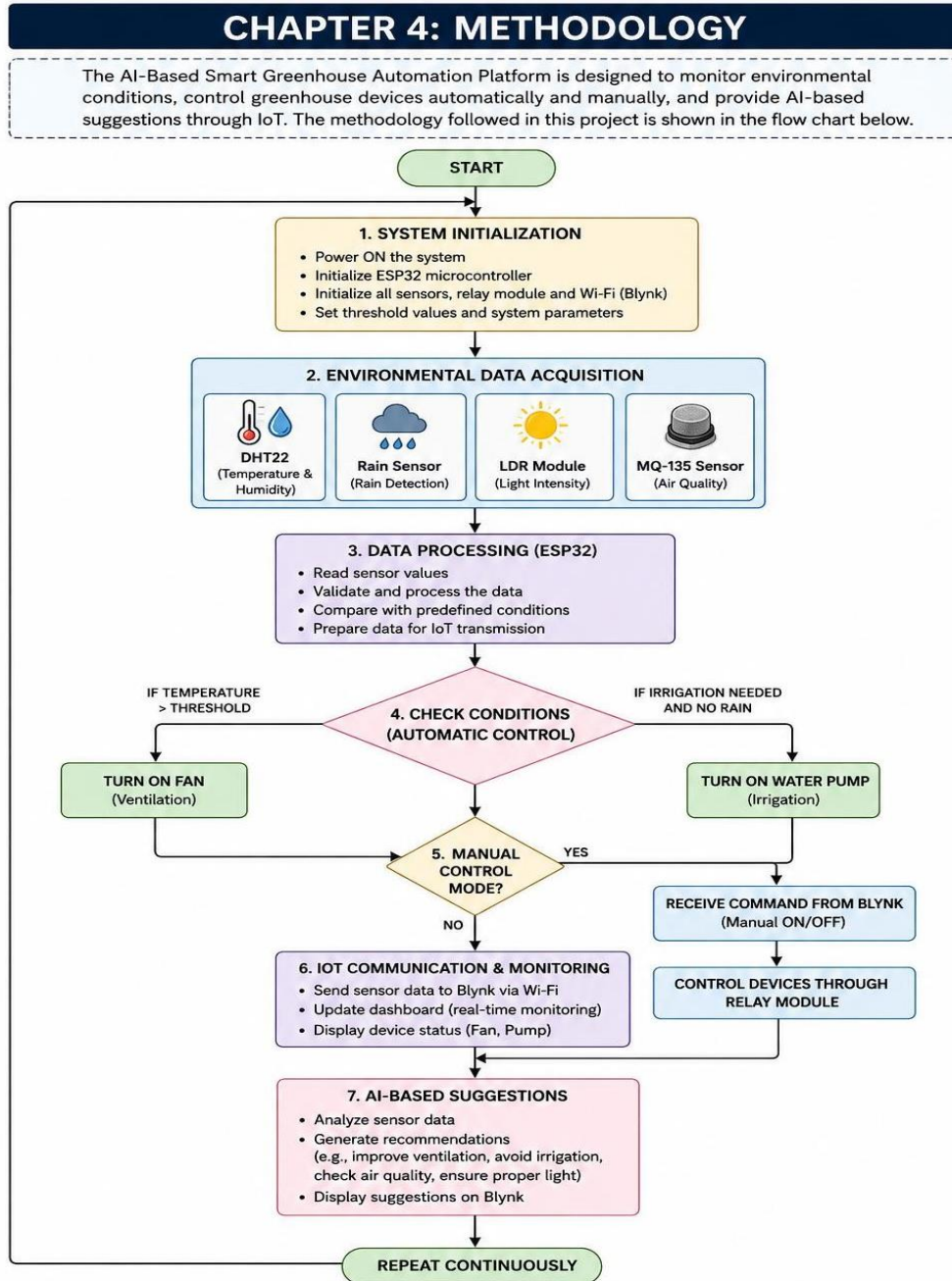


FIG 4.1 Methodolgy

4.1 SYSTEM METHODOLOGY

The proposed AI-Based Smart Greenhouse Automation Platform follows an integrated methodology that combines environmental sensing, ESP32-based processing, IoT monitoring, automatic control, manual control, and AI-based recommendations. The methodology begins with collecting greenhouse environmental parameters using sensors. The DHT22 measures temperature and humidity, the rain sensor detects rain conditions, the LDR module measures light intensity, and the MQ-135 monitors air-quality-related conditions.

The collected sensor data is received and processed by the ESP32 microcontroller. The ESP32 evaluates the sensor values according to predefined conditions and sends the real-time data to the Blynk IoT platform through Wi-Fi. The Blynk dashboard allows the user to monitor the greenhouse conditions and device status remotely. In automatic mode, the ESP32 controls the 12 V DC fan and water pump through the 4-channel relay module based on the programmed conditions. In manual mode, the user can control the connected devices through the Blynk dashboard.

The sensor values available through Blynk are also used by the AI-based recommendation system to provide meaningful suggestions to the user. Finally, the system continuously monitors the greenhouse and assists in maintaining suitable environmental conditions.

4.2 SENSOR DATA ACQUISITION

The first stage of the methodology is environmental data acquisition. Multiple sensors are installed in the greenhouse to collect important environmental parameters. The DHT22 provides temperature and humidity readings, while the rain sensor detects rainfall conditions. The LDR module measures the available light intensity, and the MQ-135 provides air-quality-related sensor readings. These sensor outputs are connected to the ESP32, which periodically reads the values and uses them for monitoring and control.

4.3 DATA PROCESSING USING ESP32

The ESP32 acts as the central controller of the system. It receives the sensor readings and processes them according to the programmed conditions. The controller determines whether the greenhouse conditions require operation of the fan or water pump. The ESP32 also manages communication with the Blynk IoT platform through Wi-Fi, allowing the sensor values and device status to be displayed in real time.

4.4 AUTOMATIC CONTROL

In automatic mode, the system compares the measured sensor values with predefined threshold conditions. When the required condition is detected, the ESP32 activates the corresponding relay.

The relay module acts as an interface between the ESP32 and the connected devices. It controls the 12 V DC fan and water pump according to the control logic. This reduces the need for continuous manual operation.

4.5 MANUAL CONTROL

The system also provides manual control through the Blynk dashboard. The user can manually switch the connected devices ON or OFF when required. Manual mode provides flexibility when the user wants to override the automatic operation for testing, maintenance, or specific greenhouse requirements.

4.6 REAL-TIME IOT MONITORING

The ESP32 connects to the Blynk IoT platform using Wi-Fi. Sensor values such as temperature, humidity, rain condition, light intensity, and air-quality-related readings are displayed on the Blynk dashboard.

The user can monitor the greenhouse remotely and observe the operating status of the fan and water pump. This provides convenient real-time access to greenhouse information.

4.7 AI-BASED RECOMMENDATION SYSTEM

The AI component uses the available sensor values displayed through the Blynk platform to provide recommendations. The purpose of the AI component is to support the user in interpreting environmental conditions.

For example:

- High temperature → Suggest increasing ventilation.
- High humidity → Suggest improving ventilation or monitoring moisture conditions.
- Rain detected → Suggest avoiding unnecessary irrigation.
- Low light → Suggest monitoring the light condition.
- Poor air-quality reading → Suggest improving ventilation and checking greenhouse conditions.

The AI component functions as a decision-support and recommendation system and does not require a separate machine-learning model.

4.8 OVERALL SYSTEM OPERATION

The complete system operates continuously by collecting environmental data, processing the readings through the ESP32, transmitting the information to Blynk, and controlling the connected devices according to predefined conditions.

The automatic mode provides automatic operation of the fan and water pump, while manual mode allows direct user control. At the same time, AI-based recommendations provide additional decision support based on the monitored sensor values.

Thus, the methodology integrates sensing, processing, IoT monitoring, automatic control, manual control, and AI-based recommendations into a single smart greenhouse automation platform.

CHAPTER 5

COMPONENT SELECTION

5.1 ESP32 MICROCONTROLLER

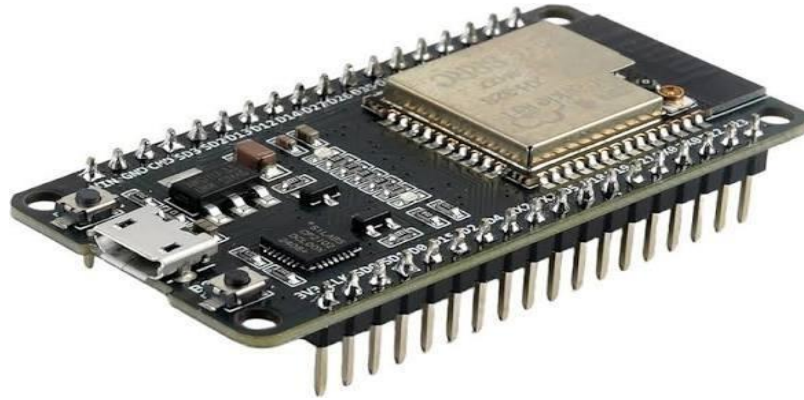


FIG 5.1 ESP32 Microcontroller

The ESP32 is the main microcontroller used in the proposed AI-Based Smart Greenhouse Automation Platform. It acts as the central processing unit of the system and is responsible for collecting data from the connected sensors, processing the sensor values, controlling the output devices, and communicating with the Blynk IoT platform. The ESP32 is particularly suitable for this project because it has built-in Wi-Fi connectivity, which allows the controller to communicate with the Blynk platform without requiring an additional communication module. It also provides multiple GPIO pins for connecting different sensors and output devices.

In the proposed system, the ESP32 receives temperature and humidity information from the DHT22 sensor, rain condition information from the rain sensor, light intensity information from the LDR module, and air-quality-related readings from the MQ-135 sensor. After receiving the sensor data, the ESP32 processes the values according to the programmed conditions. Based on these conditions, it operates the 12 V DC fan and water pump through the relay module. The ESP32 also sends the sensor readings and device status to the Blynk

dashboard through Wi-Fi, allowing the user to monitor the greenhouse remotely. Therefore, the ESP32 plays a central role in sensing, processing, communication, and automation.



The 4-channel relay module is used as a switching interface between the ESP32 and the electrical output devices. The GPIO pins of the ESP32 are designed to provide control signals and cannot directly operate devices that require higher current. Therefore, the relay module is used to safely switch the connected loads according to the control signals received from the ESP32.

When the ESP32 determines that a particular device needs to operate, it sends a control signal to the corresponding relay channel. The relay then switches

the connected device ON or OFF. This arrangement allows automatic and manual control of the greenhouse devices and provides a convenient interface between the low-power controller and the output loads.

5.3 12 V FAN



FIG 5.3 12 V Fan

The 12 V DC fan is used as the ventilation device in the proposed greenhouse system. Temperature is one of the most important environmental parameters affecting plant growth, and excessive temperature inside a greenhouse can create unsuitable conditions for plants. The fan helps improve air circulation and supports the reduction of excessive heat inside the greenhouse.

The fan is controlled by the ESP32 through the relay module. The DHT22 continuously provides temperature readings to the controller. When the measured temperature reaches the predefined condition, the ESP32 activates the corresponding relay and switches ON the fan. When the temperature returns to the desired range, the fan can be switched OFF according to the programmed control logic. The fan can also be operated manually through the Blynk dashboard when manual mode is selected. Thus, the 12 V DC fan provides an effective method of maintaining proper ventilation in the greenhouse.

5.4 WATER PUMP



FIG 5.4 Water Pump

The water pump is used to supply water to the plants and perform the irrigation function in the proposed greenhouse system. Proper irrigation is essential for healthy plant growth, but unnecessary operation of the pump can result in water wastage and unnecessary energy consumption. Therefore, automatic control of the water pump is an important part of the proposed system.

The water pump is connected to the relay module and controlled by the ESP32. The controller evaluates the programmed conditions and determines whether the pump should be operated. The rain sensor provides information about rainfall conditions, which can be considered while controlling the pump to avoid unnecessary irrigation during rain. The user can also manually switch the water pump ON or OFF using the Blynk dashboard. This combination of automatic and manual control provides flexibility in greenhouse irrigation management.

5.5 RAIN SENSOR



FIG 5.5 Rain Sensor

The rain sensor is used to detect the presence of rain or water droplets in the surrounding environment. It generally consists of a sensing plate and an electronic circuit that produces an output according to the detected wet condition. The sensor output is connected to the ESP32, which reads the signal and determines whether rain is present.

Rain detection is useful for greenhouse irrigation management because the water pump may not need to operate when sufficient rainfall is detected. The ESP32 can use the rain sensor reading as one of the conditions for automatic pump control. The rain condition can also be displayed on the Blynk dashboard so that the user can monitor it remotely. The sensor value can further be considered by the AI-based recommendation system for providing suggestions related to irrigation.

5.6 DHT22



FIG 5.6 DHT22

The DHT22 is a digital temperature and humidity sensor used to monitor the environmental conditions inside the greenhouse. Temperature and relative humidity are important factors that influence plant growth and greenhouse environmental quality. The DHT22 is capable of providing both temperature and humidity measurements using a single sensor, making it suitable for the proposed system.

The DHT22 is connected to the ESP32, which periodically reads the sensor output. The measured temperature and humidity values are processed by the ESP32 and transmitted to the Blynk IoT platform through Wi-Fi. Temperature information is also used for automatic fan control based on predefined conditions. The humidity information provides additional environmental information to the user and can be used by the AI-based recommendation system to provide suitable suggestions. The DHT22 was selected because it is simple to interface, provides digital output, and is suitable for environmental monitoring applications.

5.7 LDR MODULE



FIG 5.7 LDR Module

The LDR module is used to monitor the light intensity inside the greenhouse. LDR stands for Light Dependent Resistor, and its electrical resistance changes according to the intensity of light falling on its surface. The variation in resistance produces a corresponding change in the sensor output, which can be read by the ESP32.

Light intensity is an important environmental parameter because plants require suitable light for photosynthesis and healthy growth. The LDR module allows the system to identify changes in the available light condition and display the corresponding sensor readings through the Blynk dashboard. The monitored light condition can also be considered by the AI-based recommendation system. The LDR module was selected because it is low-cost, simple to use, and easy to interface with the ESP32.

5.8 MQ-135



FIG 5.8 MQ-135

The MQ-135 is a gas sensor used in the proposed system for monitoring air-quality-related conditions inside the greenhouse. The sensor produces an electrical output that varies according to changes in the surrounding gas environment. The ESP32 reads this sensor output and provides the corresponding information through the Blynk IoT platform.

Air quality is an important environmental factor that can influence the overall greenhouse environment. The MQ-135 provides an additional parameter for identifying changes in air conditions. The sensor reading can also support the operation of the ventilation system and provide information for AI-based recommendations. For example, when the air-quality-related reading indicates an undesirable condition, the system can suggest improving ventilation and checking the greenhouse environment. The MQ-135 was selected because it is a commonly used, low-cost gas sensor that can be integrated with microcontroller-based monitoring systems.

CHAPTER 6

SYSTEM OPERATION

SYSTEM OPERATION FLOWCHART

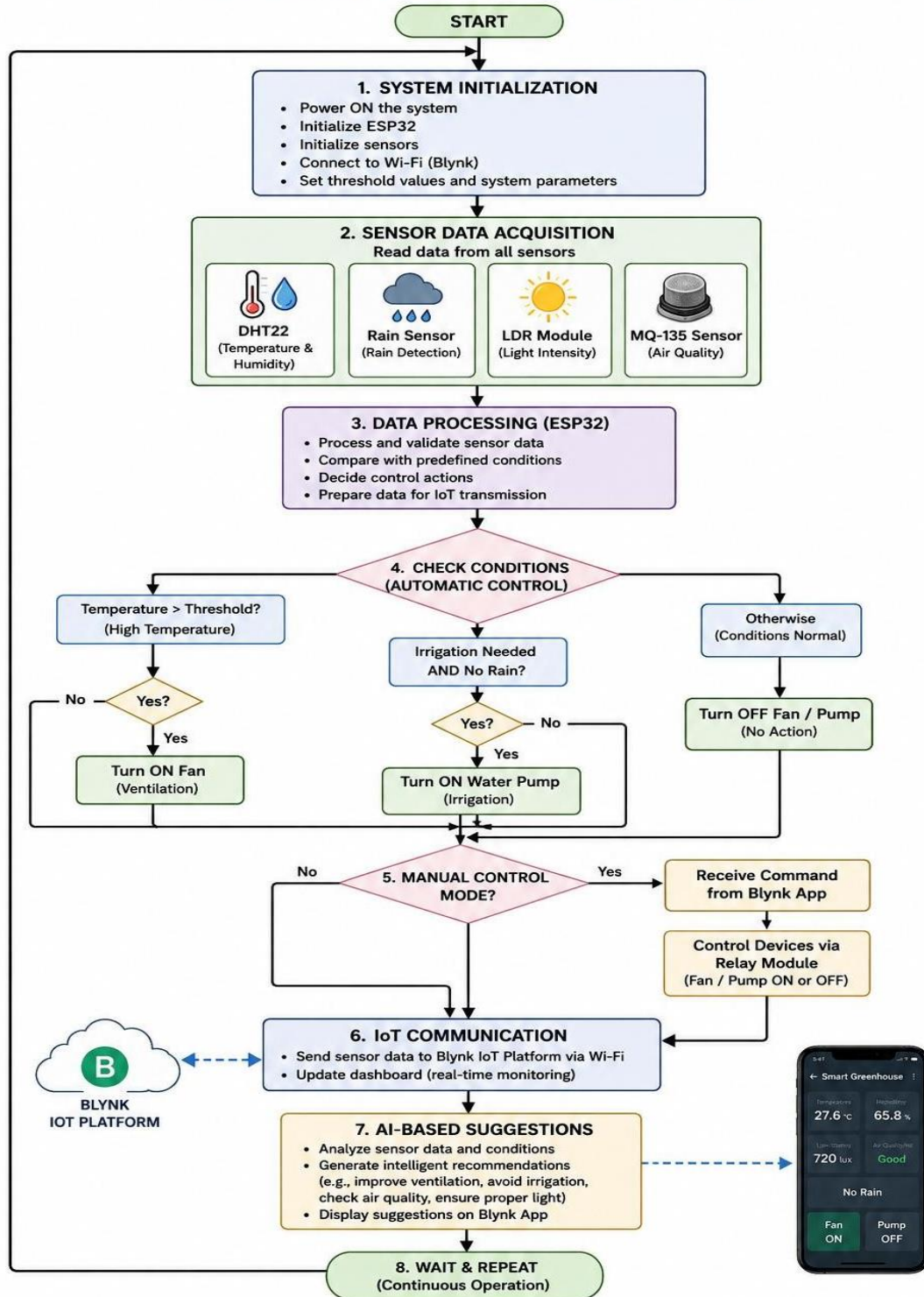


FIG 6.1 System Operation

6.1 WORKING

The AI-Based Smart Greenhouse Automation Platform is designed to continuously monitor the environmental conditions of a greenhouse and provide automatic as well as manual control of selected devices. The complete system is based on the integration of environmental sensors, an ESP32 microcontroller, a 4-channel relay module, a 12 V DC fan, a water pump, Wi-Fi communication, the Blynk IoT platform, and an AI-based recommendation system. The sensors collect information from the greenhouse and provide the data to the ESP32. The controller processes the received values according to the programmed conditions and performs the required control actions.

The DHT22 sensor is used to measure temperature and humidity, while the rain sensor is used to identify rain conditions. The LDR module provides information about the light intensity, and the MQ-135 sensor monitors air-quality-related conditions. The ESP32 receives these sensor outputs and communicates with the Blynk IoT platform using its built-in Wi-Fi capability. The sensor values are displayed on the Blynk dashboard, allowing the user to monitor the greenhouse conditions remotely.

The system operates in both automatic and manual modes. In automatic mode, the ESP32 evaluates the sensor values and controls the fan and water pump through the relay module according to the predefined conditions. In manual mode, the user can control the connected devices through the Blynk dashboard. The AI component uses the available sensor values to provide meaningful suggestions to the user. Therefore, the complete operation combines sensing, processing, communication, automation, manual control, and intelligent decision support.

6.1.1 Environmental Data Acquisition

The first stage of the system operation is environmental data acquisition. The sensors are positioned inside or around the greenhouse to obtain information about the surrounding environmental conditions. The DHT22 sensor continuously measures temperature and relative humidity. The rain sensor detects the presence of rain or water droplets, while the LDR module detects changes in light intensity. The MQ-135 sensor provides information related to the air-quality condition.

The sensor outputs are connected to the appropriate input pins of the ESP32. The microcontroller periodically reads the sensor values and converts them into usable information for further processing. The frequency of data collection can be determined by the program depending on the requirements of the greenhouse monitoring application.

The collected sensor values form the basic input for the entire automation system. Instead of depending on manual observation, the system continuously obtains environmental information and uses the data for monitoring and control. This provides a consistent method of observing changes in the greenhouse environment.

6.1.2 Temperature and Humidity Monitoring

Temperature and humidity monitoring is an important function because both parameters have a direct influence on the greenhouse environment and plant growth. The DHT22 sensor is used to measure these parameters continuously. The sensor sends the measured values to the ESP32, which processes and transmits them to the Blynk IoT platform.

The temperature value is used as an important input for the ventilation control. When the measured temperature reaches the predefined condition, the ESP32 activates the relay connected to the 12 V DC fan. The fan provides air circulation and assists in reducing excessive heat inside the greenhouse. Once the environmental condition returns to the desired range, the controller can switch OFF the fan according to the programmed logic.

Humidity is monitored along with temperature because excessive or insufficient humidity can indicate an undesirable greenhouse condition. The humidity value is displayed on the Blynk dashboard so that the user can observe the environmental condition remotely. Both temperature and humidity values can also be considered by the AI-based recommendation system for providing suitable suggestions.

6.1.3 Rain Condition Detection

The rain sensor is used to identify the occurrence of rain or the presence of water on its sensing surface. When water droplets come into contact with the sensing plate, the electrical output of the sensor changes. This output is read by the ESP32 and processed to determine the rain condition.

Rain detection is particularly useful for irrigation management. When rain is detected, unnecessary operation of the water pump can be avoided according to the programmed control logic. This can help reduce water wastage and unnecessary energy consumption. The detected rain condition is also communicated to the Blynk platform, allowing the user to monitor the condition remotely.

The rain information can also be used by the AI-based recommendation system. For example, when rain is detected, the system can recommend avoiding unnecessary irrigation. This allows the rain sensor to contribute not only to automatic control but also to intelligent decision support.

6.1.4 Light Intensity and Air Quality Monitoring

The LDR module is used to monitor the light condition inside the greenhouse. The resistance of the Light Dependent Resistor changes according to the intensity of light falling on its surface. The resulting output is read by the ESP32 and processed as a light-related sensor value.

Light is an essential factor for plant growth because plants require sufficient light for photosynthesis. Continuous monitoring of light intensity allows the user to understand the changing light conditions throughout the day. The sensor value is transmitted to the Blynk dashboard and can also be used by the AI-based recommendation system to provide suggestions when an undesirable light condition is observed.

The MQ-135 sensor is used to monitor air-quality-related conditions. The sensor produces an output that changes according to variations in the surrounding gas environment. The ESP32 reads this output and sends the information to the Blynk platform. The air-quality-related reading provides an additional indication of the greenhouse environment.

If the monitored air-quality-related condition becomes undesirable, the AI-based recommendation system can suggest improving ventilation or checking the greenhouse environment. In this way, the LDR and MQ-135 provide additional environmental information beyond temperature, humidity, and rain.

6.1.5 Automatic Fan and Water Pump Control

Automatic control is one of the main features of the proposed system. The ESP32 uses predefined conditions to determine when the fan or water pump should be operated. The controller continuously evaluates the sensor values and generates the required control signals.

The 12 V DC fan is controlled based mainly on the temperature condition. When the temperature exceeds the programmed threshold, the ESP32 sends a signal to the appropriate relay channel. The relay switches ON and supplies the required connection to operate the fan. The fan then provides ventilation inside the greenhouse.

The water pump is controlled according to the programmed irrigation conditions. Rain detection can be considered in the control process so that the pump does not operate unnecessarily during detected rain conditions. The pump is switched through the relay module, allowing the ESP32 to control a higher-load device without directly driving it from the microcontroller GPIO.

The automatic control process reduces continuous human involvement. Instead of manually observing the greenhouse and operating the devices, the system responds to the programmed environmental conditions. This improves convenience and provides a consistent method of device operation.

6.1.6 Manual Device Control

The proposed system also provides a manual operating mode to give the user direct control over the greenhouse devices. Manual control is useful when the user wants to operate the fan or water pump independently of the automatic control conditions.

The user can access the Blynk dashboard through a connected device and select the required control option. The command is transmitted through the internet to the ESP32. The ESP32 receives the command and controls the appropriate relay channel. The relay then switches the selected device ON or OFF.

Manual control is also useful during system testing and maintenance. For example, the user can manually switch ON the fan to verify its operation or activate the water pump when irrigation is required. Thus, the manual mode provides flexibility and allows the user to take direct control whenever necessary.

6.1.7 Real-Time IoT Monitoring and AI-Based Suggestions

The ESP32 communicates with the Blynk IoT platform through Wi-Fi to provide real-time monitoring of the greenhouse. Sensor readings collected from the DHT22, rain sensor, LDR module, and MQ-135 are transmitted to the Blynk dashboard. The user can observe the environmental values without being physically present near the greenhouse.

The dashboard can display temperature, humidity, rain condition, light intensity, and air-quality-related readings along with the status of the connected devices. This provides a centralized interface for monitoring the greenhouse and observing the operation of the fan and water pump.

The AI component uses the sensor values available through the Blynk dashboard as the basis for recommendations. It does not require a separate machine-learning model in the proposed implementation. Instead, it acts as a decision-support system that interprets the available sensor conditions and provides useful suggestions.

For example, when the temperature is high, the system can suggest increasing ventilation. When rain is detected, it can suggest avoiding unnecessary irrigation. When an undesirable air-quality-related condition is observed, it can suggest improving air circulation. Similarly, changes in light and humidity can be highlighted to the user for further observation.

6.1.8 Overall System Performance

The overall performance of the proposed system depends on the coordinated operation of all hardware and software components. The DHT22, rain sensor, LDR module, and MQ-135 continuously provide environmental information to the ESP32. The ESP32 processes these readings and communicates the information to the Blynk platform through Wi-Fi.

The automatic control system responds to predefined environmental conditions by operating the 12 V DC fan and water pump through the relay module. The manual mode allows the user to override or independently operate the devices when required. The Blynk platform provides real-time visibility of sensor readings and device status, while the AI-based recommendation system provides additional guidance based on the available sensor information.

The integration of these functions reduces the requirement for continuous manual monitoring and allows the greenhouse environment to be observed and managed more conveniently. The modular design also makes the system suitable for future expansion. Additional sensors, actuators, data storage, advanced prediction techniques, and improved AI capabilities can be integrated into the existing platform.

Overall, the proposed system provides an integrated approach to greenhouse management by combining environmental sensing, ESP32-based processing, IoT connectivity, automatic control, manual operation, and AI-based recommendations. This makes the platform a practical and expandable solution for smart greenhouse monitoring and automation.

CHAPTER 7

CONCLUSION

The AI-Based Smart Greenhouse Automation Platform provides an effective solution for monitoring and controlling greenhouse environmental conditions. The system uses ESP32, DHT22, rain sensor, LDR module, and MQ-135 to monitor temperature, humidity, rain, light intensity, and air-quality-related conditions. The collected sensor data is displayed on the Blynk IoT platform for real-time remote monitoring. The 12 V DC fan and water pump are automatically controlled through a relay module based on predefined conditions, while manual control through Blynk provides additional flexibility to the user.

The AI-based recommendation system uses the available sensor values to provide useful suggestions for greenhouse management. The system also improves the convenience of greenhouse management by combining automatic operation, manual control, real-time monitoring, and AI-assisted decision support in a single platform. By continuously observing environmental conditions and responding to changes, the system helps maintain a suitable environment for plant growth while reducing unnecessary operation of the fan and water pump. The modular design also makes it easy to add new sensors and control devices in the future. Overall, the proposed system reduces manual intervention and provides a low-cost, practical, reliable, and expandable solution for smart agriculture and greenhouse management.

FUTURE SCOPE

The proposed system has considerable scope for future development and improvement. One of the major improvements can be the integration of a soil moisture sensor. The present system monitors environmental parameters such as temperature, humidity, rain, light intensity, and air quality, but direct measurement of soil moisture would provide more accurate information about the actual water requirement of plants. The water pump could then be controlled based on soil moisture along with the other environmental conditions.

The AI component can also be developed further by incorporating machine-learning algorithms. At present, the AI component provides recommendations based on the available sensor values. In the future, historical sensor data can be collected and stored in a database. Machine-learning models can then be trained using this historical information to identify environmental patterns and predict future greenhouse conditions. This could enable the system to provide more advanced recommendations for irrigation, ventilation, and environmental management.

Future versions can also include crop-specific recommendations. Different plants require different temperature, humidity, light, and irrigation conditions. A crop-selection feature could allow the user to select the plant being cultivated, after which the system could provide recommendations based on the specific requirements of that crop. This would make the system more useful for practical agricultural applications.

Another possible improvement is the addition of automatic notification and alert functions. The system could send notifications to the user when temperature becomes excessively high, humidity reaches an undesirable level, rain is detected, air-quality-related conditions become unsuitable, or a connected

device fails to operate. Such notifications could help the user respond quickly to important changes in the greenhouse environment.

The system can also be expanded to control additional greenhouse equipment. Future versions could include automated lighting, cooling systems, misting systems, exhaust fans, fertilizer pumps, and other agricultural equipment. Multiple relay channels can be utilized to control additional devices. This would allow the system to progress from basic monitoring and control toward a more comprehensive greenhouse automation platform.

Solar power integration is another important area for future development. Solar panels and a suitable battery system could be used to supply power to the sensors, ESP32, and other low-power components. This could reduce dependence on conventional electrical sources and improve the suitability of the system for remote agricultural locations.

Cloud-based data storage can also be introduced to maintain long-term records of greenhouse conditions. The stored data could be displayed using graphs and statistical analysis to help users understand environmental changes over time. This information could also be used to evaluate the performance of the greenhouse and improve future control strategies.

The system can further be developed into a multi-zone greenhouse monitoring platform. Multiple ESP32 controllers or sensor nodes could be installed in different sections of a large greenhouse. Each zone could independently monitor temperature, humidity, light, and air-quality-related conditions. A centralized dashboard could then provide information from all zones and allow individual control of devices.

In the future, advanced AI techniques can be combined with weather information and historical greenhouse data to improve prediction and decision-making. The system could predict environmental changes and take preventive action before unsuitable conditions occur. With these improvements, the proposed platform can be transformed into a more intelligent, autonomous, energy-efficient, and scalable agricultural automation system.

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PROTOTYPE PHOTOGRAPHIC